

A Hybrid Second Moment Method for Thermal Radiative Transfer

M&C 2025

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April 29, 2025



LLNL-PRES-2005149

This work was performed under the auspices of the U.S. Department of Energy by Lawrence Livermore National Laboratory under contract DE-AC52-07NA27344. Lawrence Livermore National Security, LLC

We describe a new hybrid method that did not immediately work.

Motivation

Prior Work

Hybrid Second Moment Method

Deterministic Component of HSM

Monte Carlo Component of HSM

Numerical Results

Method of Manufactured Solutions Verification

Noisy Crooked Pipe

Variance Reduction

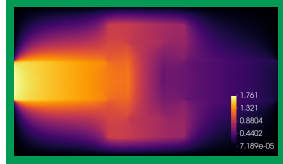
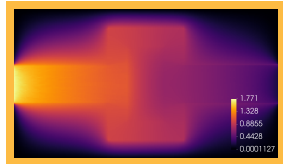
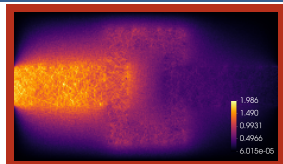
Difference Formulation

Incorrect Crooked Pipe

Generalized Difference Formulation

Correct Crooked Pipe

Conclusion



Can we make Monte Carlo faster without sacrificing accuracy?

- Why Monte Carlo?
- Why IMC?
- Why not RW?
- Why not DDMC?
- Why not Cooper, Lam, Novellino, Park, Pasmann, Willert, or any other hybrid methods?

Progress Porting LLNL Monte Carlo Transport Codes to the AMD Instinct MI300A APU at 1:25 PM today in Special Session on GPU Computing



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Hybrid Second Moment (HSM) solves a linear transport equation.

$$\boldsymbol{\Omega} \cdot \nabla \psi + \sigma_t \psi = \frac{\sigma_s}{4\pi} \int_{\mathbb{S}^2} \psi \, d\Omega' + q, \quad (1a)$$

$$\psi(\mathbf{x}, \boldsymbol{\Omega}) = \psi_{\text{inc}}(\mathbf{x}, \boldsymbol{\Omega}), \quad \mathbf{x} \in \partial\mathcal{D} \text{ and } \boldsymbol{\Omega} \cdot \mathbf{n} < 0. \quad (1b)$$

This system is an equivalent reformulation of eqs. (1a) and (1b).

$$\nabla \cdot \mathbf{J} + \sigma_a \varphi = Q_0, \quad \mathbf{x} \in \mathcal{D}, \quad (2a)$$

$$\frac{1}{3} \nabla \varphi + \sigma_t \mathbf{J} = \mathbf{Q}_1 - \nabla \cdot \mathbf{T}, \quad \mathbf{x} \in \mathcal{D}. \quad (2b)$$

$$\mathbf{J} \cdot \mathbf{n} = \frac{1}{2} \varphi + 2J_{\text{in}} + \beta, \quad \mathbf{x} \in \partial \mathcal{D}. \quad (2c)$$

The Second Moment Method iterates the scattering source.

$$\begin{array}{lcl}
 & \varphi & \\
 \Omega \cdot \nabla \psi + \sigma_t \psi = \frac{\sigma_s}{4\pi} \varphi + q, & \nabla \cdot \mathbf{J} + \sigma_a \varphi = Q_0, & \mathbf{x} \in \mathcal{D}, \\
 \psi(\mathbf{x}, \Omega) = \bar{\psi}(\mathbf{x}, \Omega), & \frac{1}{3} \nabla \varphi + \sigma_t \mathbf{J} = \mathbf{Q}_1 - \nabla \cdot \mathbf{T}, & \mathbf{x} \in \mathcal{D}, \\
 \mathbf{x} \in \partial \mathcal{D} \text{ and } \Omega \cdot \mathbf{n} < 0. & \mathbf{J} \cdot \mathbf{n} = \frac{1}{2} \varphi + 2J_{\text{in}} + \beta, & \mathbf{x} \in \partial \mathcal{D}. \\
 & & \\
 \mathbf{T}(\psi) = \int_{\mathbb{S}^2} \Omega \otimes \Omega \psi \, d\Omega - \frac{1}{3} \mathbf{I} \int_{\mathbb{S}^2} \psi \, d\Omega & & \\
 \beta(\psi) = \int_{\mathbb{S}^2} |\Omega \cdot \mathbf{n}| \psi \, d\Omega - \frac{1}{2} \int_{\mathbb{S}^2} \psi \, d\Omega & &
 \end{array}$$

The deterministic component of HSM solves the moment system.

Find $(\varphi, \mathbf{J}) \in Y_p \times RT_p$ such that,

$$\int u \nabla \cdot \mathbf{J} \, d\mathbf{x} + \int \sigma_a u \varphi \, d\mathbf{x} = \int u Q_0 \, d\mathbf{x}, \quad \forall u \in Y_p, \quad (3a)$$

$$\begin{aligned} & -\frac{1}{3} \int \nabla \cdot \mathbf{v} \varphi \, d\mathbf{x} + \int \sigma_t \mathbf{v} \cdot \mathbf{J} \, d\mathbf{x} + \frac{2}{3} \int_{\Gamma_b} (\mathbf{v} \cdot \mathbf{n})(\mathbf{J} \cdot \mathbf{n}) \, ds = \int \mathbf{v} \cdot \mathbf{Q}_1 \, d\mathbf{x} - \int_{\Gamma_b} \mathbf{v} \cdot \mathbf{Tn} \, ds \\ & + \frac{2}{3} \int_{\Gamma_b} (\mathbf{v} \cdot \mathbf{n})(2J_{\text{in}} + \beta) \, ds - \int_{\Gamma_0} \llbracket v \rrbracket \cdot \{\{\mathbf{Tn}\}\} \, ds + \int \nabla_h \mathbf{v} : \mathbf{T} \, d\mathbf{x} \quad \forall \mathbf{v} \in RT_p. \end{aligned} \quad (3b)$$

The Monte Carlo component of HSM has no scattering events.

Estimate $\hat{\mathbf{T}} = \hat{\mathbf{P}} - \frac{1}{3}\hat{\mathbf{I}}\hat{\phi}$ and $\hat{\beta} = \hat{B} - \frac{1}{2}\hat{\phi}_s$ using:

$$\hat{\phi} = \frac{1}{V} \sum_i d_i w_i, \quad \hat{\mathbf{P}} = \frac{1}{V} \sum_i \boldsymbol{\Omega}_i \otimes \boldsymbol{\Omega}_i d_i w_i, \quad (4a)$$

$$\hat{B} = \frac{2}{A} \sum_i w_i, \quad \hat{\phi}_s = \frac{2}{A} \sum_i \frac{w_i}{|\boldsymbol{\Omega}_i \cdot \mathbf{n}|}. \quad (4b)$$

While solving:

$$\boldsymbol{\Omega} \cdot \nabla \psi + \sigma_t \psi = \frac{\sigma_s}{4\pi} \varphi + q, \quad (5a)$$

$$\psi(\mathbf{x}, \boldsymbol{\Omega}) = \psi_{\text{inc}}(\mathbf{x}, \boldsymbol{\Omega}), \quad \mathbf{x} \in \partial\mathcal{D} \text{ and } \boldsymbol{\Omega} \cdot \mathbf{n} < 0. \quad (5b)$$

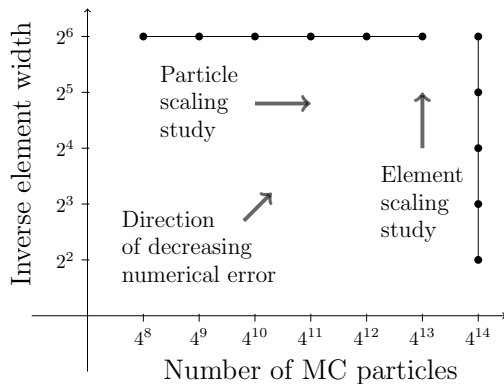
Using MC allows us to move fixed sources out of the iteration.

```
1:  $\hat{\phi}^{(0)}, \hat{\mathbf{T}}^{(0)}, \hat{\beta}^{(0)} \leftarrow \text{mc}(q, \psi_{\text{inc}})$ 
2:  $i \leftarrow 1$ 
3: while not converged( $\hat{\phi}^{(i-1)}, \hat{\phi}^{(i)}$ ) do
4:    $\varphi^{(i)} \leftarrow \text{sm}(Q_0, Q_1, \hat{\mathbf{T}}^{(i-1)}, \hat{\beta}^{(i-1)})$ 
5:    $\hat{\phi}_{\text{temp}}, \hat{\mathbf{T}}_{\text{temp}}, \hat{\beta}_{\text{temp}} \leftarrow \text{mc}(\varphi^{(i)})$ 
6:    $\hat{\phi}^{(i)} \leftarrow \hat{\phi}^{(0)} + \hat{\phi}_{\text{temp}}$ 
7:    $\hat{\mathbf{T}}^{(i)} \leftarrow \hat{\mathbf{T}}^{(0)} + \hat{\mathbf{T}}_{\text{temp}}$ 
8:    $\hat{\beta}^{(i)} \leftarrow \hat{\beta}^{(0)} + \hat{\beta}_{\text{temp}}$ 
9:    $i \leftarrow i + 1$ 
10: end while
11: return  $\hat{\phi}^{(i)}$ 
```

▷ Sample the fixed sources q and ψ_{inc}

▷ Sample the variable source $\frac{\sigma_s}{4\pi} \varphi^{(i)}$

The HSM error surface height is lowest in the north-east.



The HSM error is:

$$O(h) + O(N^{-1/2})$$

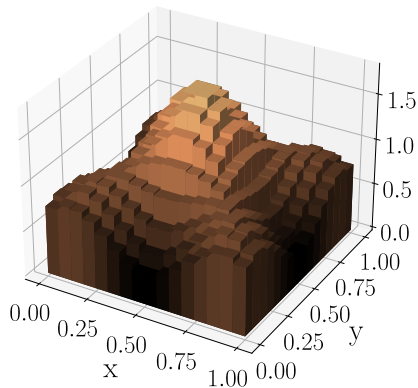
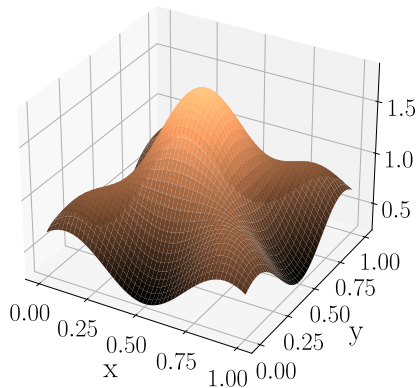
We verified HSM using the Method of Manufactured Solutions.

$$\psi_{\text{MMS}}(\mathbf{x}, \boldsymbol{\Omega}) = \frac{1}{4\pi} \left(\sin(\pi x) \sin(\pi y) + \boldsymbol{\Omega}_x \boldsymbol{\Omega}_y \sin(2\pi x) \sin(2\pi y) \right. \\ \left. + \boldsymbol{\Omega}_x^2 \sin\left(\frac{5\pi}{2}x + \frac{\pi}{4}\right) \sin\left(\frac{5\pi}{2}y + \frac{\pi}{4}\right) + 0.5 \right), \quad (6a)$$

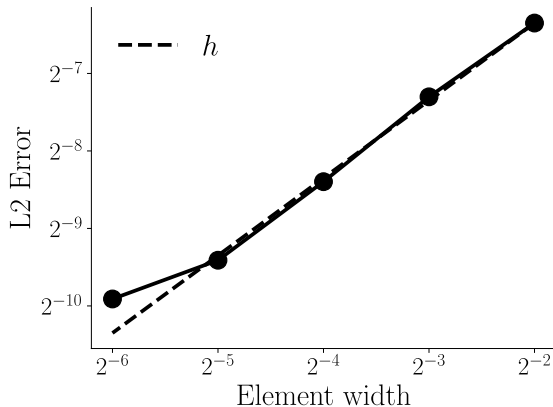
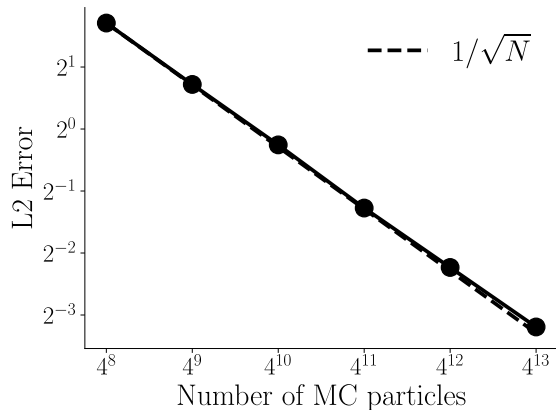
$$\phi_{\text{MMS}}(\mathbf{x}) = \int_{\mathbb{S}^2} \psi_{\text{MMS}}(\mathbf{x}, \boldsymbol{\Omega}) \, d\Omega, \quad (6b)$$

$$\bar{\phi}_{\text{MMS}}(x_1, x_2, y_1, y_2) = \frac{1}{((x_2 - x_1)(y_2 - y_1))} \int_{x_1}^{x_2} \int_{y_1}^{y_2} \phi_{\text{MMS}}(\mathbf{x}) \, dy \, dx. \quad (6c)$$

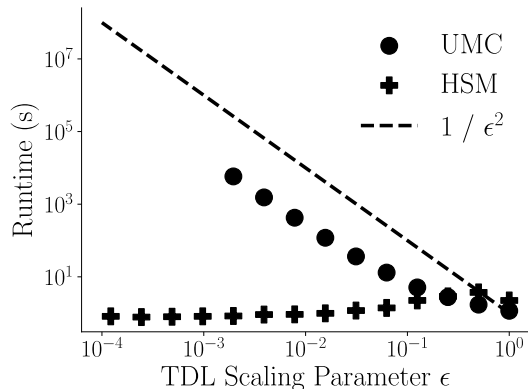
The MMS solution that we seek with HSM is a histogram.



We observed the hypothesized error for the MMS problem.



HSM is faster than unaccelerated MC (UMC) in the TDL.



Thick Diffusion Limit (TDL):

$$\sigma_t \rightarrow \sigma_t / \epsilon, \quad (7a)$$

$$\sigma_a \rightarrow \epsilon \sigma_a, \quad (7b)$$

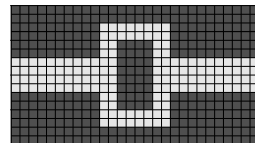
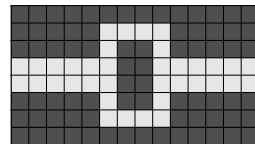
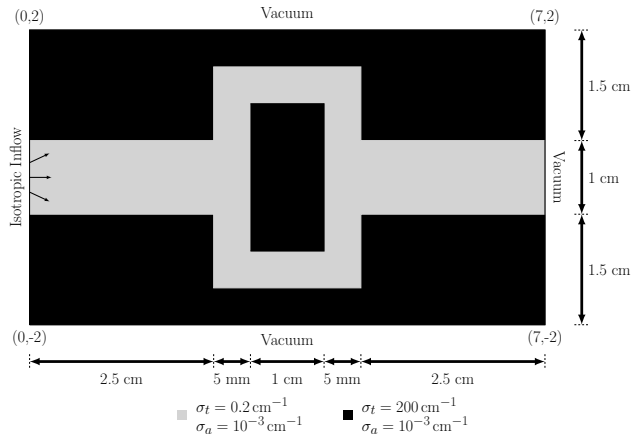
$$\sigma_s = \sigma_t - \sigma_a, \quad (7c)$$

$$q \rightarrow \epsilon q, \quad (7d)$$

for the TDL parameter $\epsilon \in (0, 1]$, and let:

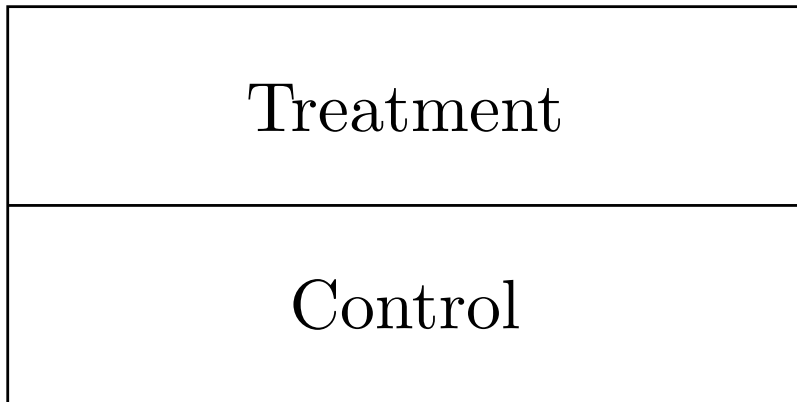
$$\epsilon \rightarrow 0. \quad (7e)$$

The Crooked Pipe is commonly used for comparing TRT methods.

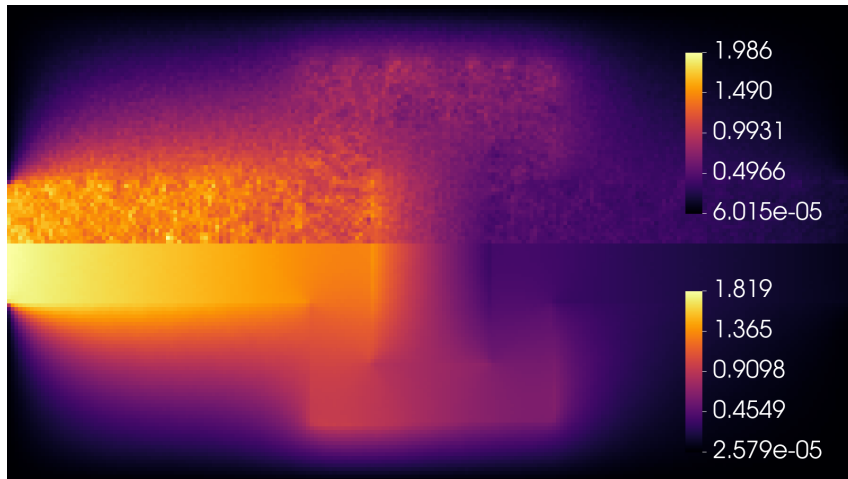


$$224 \times 128 = 28,672$$

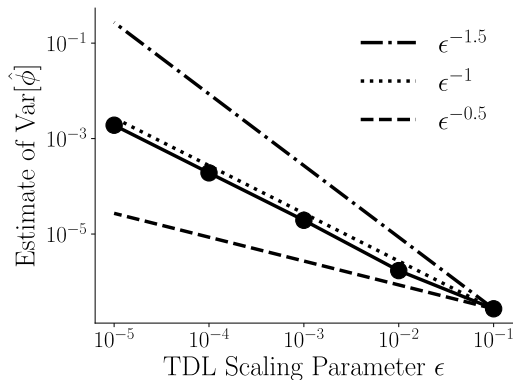
The horizontal symmetry of CP allows top/bottom plots.



HSM is much more noisy than unaccelerated Monte Carlo.



The HSM variance is very large in the Thick Diffusion Limit.



$$\text{Var}[\hat{\phi}] \approx \frac{1}{600} \sum_{i=1}^{600} (\hat{\phi}_i - \Phi)^2, \quad (8)$$

where Φ is the mean estimate defined by,

$$\Phi = \frac{1}{600} \sum_{i=1}^{600} \hat{\phi}_i. \quad (9)$$

The rest of this talk is about variance reduction.

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Variance Reduction

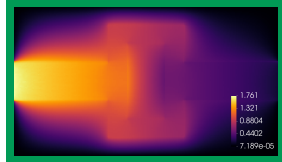
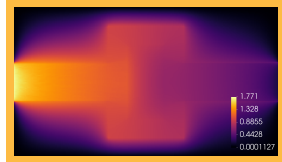
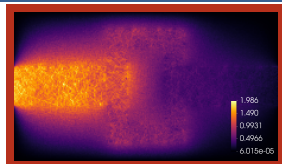
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Conclusion



Compute the deviation of the intensity from isotropy.

Substitute $\psi = \frac{\varphi}{4\pi} + \tilde{\psi}$ into:

$$\mathbf{\Omega} \cdot \nabla \psi + \sigma_t \psi = \frac{\sigma_s}{4\pi} \varphi + q, \quad (10a)$$

$$\psi(\mathbf{x}, \mathbf{\Omega}) = \psi_{\text{inc}}(\mathbf{x}, \mathbf{\Omega}), \quad \mathbf{x} \in \partial\mathcal{D} \text{ and } \mathbf{\Omega} \cdot \mathbf{n} < 0. \quad (10b)$$

to get,

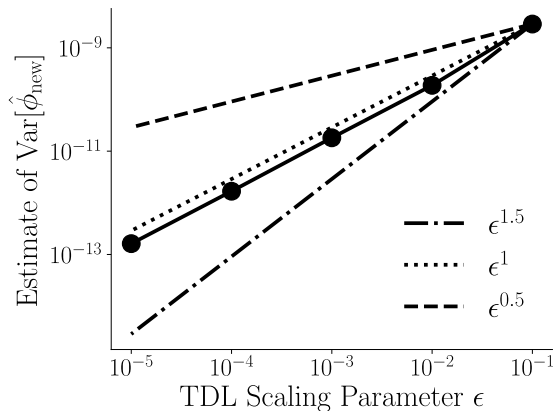
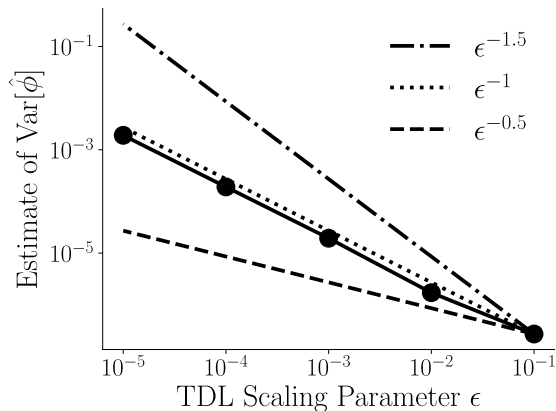
$$\mathbf{\Omega} \cdot \nabla \tilde{\psi} + \sigma_t \tilde{\psi} = -\frac{1}{4\pi} (\sigma_a \varphi + \mathbf{\Omega} \cdot \nabla \varphi) + q, \quad (11a)$$

$$\tilde{\psi}(\mathbf{x}, \mathbf{\Omega}) = \psi_{\text{inc}}(\mathbf{x}, \mathbf{\Omega}) - \frac{\varphi(\mathbf{x})}{4\pi}, \quad \mathbf{x} \in \partial\mathcal{D} \text{ and } \mathbf{\Omega} \cdot \mathbf{n} < 0. \quad (11b)$$

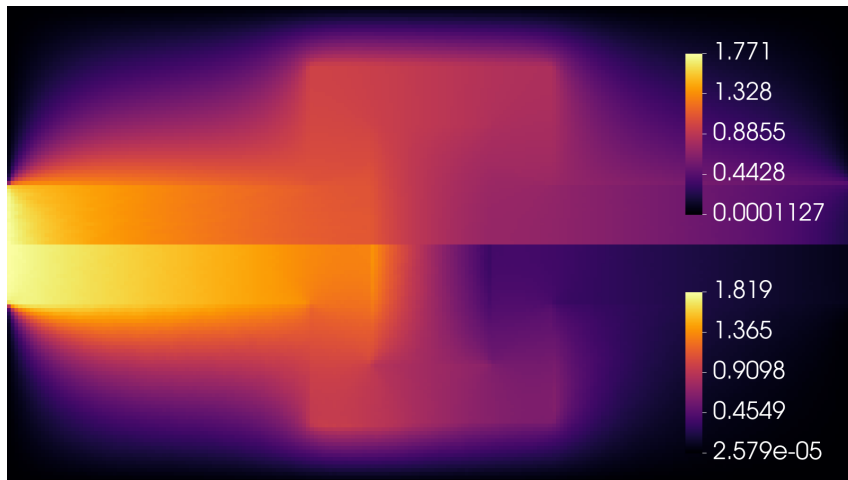
This approach requires $\nabla\varphi$ and a variable boundary source.

	Fixed	Variable		
Volume	q	$\frac{\sigma_s}{4\pi}\varphi$	$O(\epsilon)$	$O(1/\epsilon)$
Boundary	ψ_{inc}	0	$O(1)$	N/A
Volume	q	$-\frac{1}{4\pi}(\sigma_a\varphi + \mathbf{\Omega} \cdot \nabla\varphi)$	$O(\epsilon)$	$O(1)$
Boundary	ψ_{inc}	$-\frac{\varphi}{4\pi}$	$O(1)$	$O(1)$

The new variance $\text{Var}[\hat{\phi}_{\text{new}}]$ is $\ll \text{Var}[\hat{\phi}]$ in the TDL ($\epsilon \ll \epsilon^{-1}$).



The new solution is incorrect on material interfaces.



Modify the approach by replacing φ with $\bar{\varphi}$.

Before, we substituted $\psi = \frac{\varphi}{4\pi} + \tilde{\psi}$ to get:

$$\mathbf{\Omega} \cdot \nabla \tilde{\psi} + \sigma_t \tilde{\psi} = -\frac{1}{4\pi}(\sigma_a \varphi + \mathbf{\Omega} \cdot \nabla \varphi) + q, \quad (12a)$$

$$\tilde{\psi}(\mathbf{x}, \mathbf{\Omega}) = \psi_{\text{inc}}(\mathbf{x}, \mathbf{\Omega}) - \frac{\varphi(\mathbf{x})}{4\pi}, \quad \mathbf{x} \in \partial\mathcal{D} \text{ and } \mathbf{\Omega} \cdot \mathbf{n} < 0. \quad (12b)$$

Now, substitute $\psi = \frac{\bar{\varphi}}{4\pi} + \tilde{\psi}$ to get:

$$\mathbf{\Omega} \cdot \nabla \tilde{\psi} + \sigma_t \tilde{\psi} = \frac{\sigma_t}{4\pi}(\varphi - \bar{\varphi}) - \frac{1}{4\pi}(\sigma_a \varphi + \mathbf{\Omega} \cdot \nabla \bar{\varphi}) + q, \quad (13a)$$

$$\tilde{\psi}(\mathbf{x}, \mathbf{\Omega}) = \psi_{\text{inc}}(\mathbf{x}, \mathbf{\Omega}) - \frac{\bar{\varphi}(\mathbf{x})}{4\pi}, \quad \mathbf{x} \in \partial\mathcal{D} \text{ and } \mathbf{\Omega} \cdot \mathbf{n} < 0. \quad (13b)$$

Choose $\bar{\varphi}$ such that $\nabla \bar{\varphi}$ is well-defined and $\varphi - \bar{\varphi}$ is $O(1/\sigma_t)$.

	Fixed	Variable		
Volume	q	$\frac{\sigma_s}{4\pi}\varphi$	$O(\epsilon)$	$O(1/\epsilon)$
Boundary	ψ_{inc}	0	$O(1)$	N/A
Volume	q	$-\frac{1}{4\pi}(\sigma_a\varphi + \mathbf{\Omega} \cdot \nabla\varphi)$	$O(\epsilon)$	$O(1)$
Boundary	ψ_{inc}	$-\frac{\varphi}{4\pi}$	$O(1)$	$O(1)$
Volume	q	$\frac{\sigma_t}{4\pi}(\varphi - \bar{\varphi}) - \frac{1}{4\pi}(\sigma_a\varphi + \mathbf{\Omega} \cdot \nabla\bar{\varphi})$	$O(\epsilon)$	$O(?)$
Boundary	ψ_{inc}	$-\frac{\bar{\varphi}}{4\pi}$	$O(1)$	$O(1)$

Solve a transient heat conduction equation for $\bar{\varphi}$.

$$\frac{\partial \bar{\varphi}}{\partial t} = \nabla \cdot \frac{1}{\sigma_t} \nabla \bar{\varphi}, \quad \mathbf{x} \in \mathcal{D}, \quad (14a)$$

$$\bar{\varphi}(\mathbf{x}, 0) = \varphi(\mathbf{x}), \quad \mathbf{x} \in \mathcal{D}, \quad (14b)$$

$$\bar{\varphi}(\mathbf{x}, t) = \varphi(\mathbf{x}), \quad \mathbf{x} \in \partial\mathcal{D}. \quad (14c)$$

Take one small timestep such that,

$$\frac{\Delta t}{h^2} \frac{1}{\max_{\mathbf{x}} \sigma_t} \ll 1. \quad (15)$$

Discretize space using discontinuous finite elements.

Find $\bar{\varphi} \in Y_p$ such that,

$$\begin{aligned} & \frac{\partial}{\partial t} \int u \bar{\varphi} \, dx - \int_{\Gamma_b} u \left(\frac{1}{\sigma_t} \nabla \bar{\varphi} \right) \cdot \mathbf{n} \, ds + \int \nabla u \cdot \frac{1}{\sigma_t} \nabla \bar{\varphi} \, d\mathbf{x} \\ & - \int_{\Gamma_0} \left\{ \left\{ \frac{1}{\sigma_t} \nabla \bar{\varphi} \cdot \mathbf{n} \right\} \right\} \llbracket u \rrbracket \, ds + \sigma \int_{\Gamma_0} \llbracket \bar{\varphi} \rrbracket \left\{ \left\{ \frac{1}{\sigma_t} \nabla u \cdot \mathbf{n} \right\} \right\} \, ds = \\ & - \sigma \int_{\Gamma_b} \varphi \left(\frac{1}{\sigma_t} \nabla u \cdot \mathbf{n} \right) \, ds - k \int_{\Gamma_b} \left\{ \left\{ \frac{1}{h} \frac{1}{\sigma_t} \right\} \right\} \varphi u \, ds, \quad \forall u \in Y_p. \quad (16) \end{aligned}$$

Discretize time using backward Euler.

Let Φ contain L^2 dofs for $\bar{\varphi}$. Then,

$$M \frac{d\Phi}{dt} = -K\Phi + f. \quad (17)$$

Backward Euler is,

$$M \frac{\Phi^{n+1} - \Phi^n}{\Delta t} = -K\Phi^{n+1} + f. \quad (18)$$

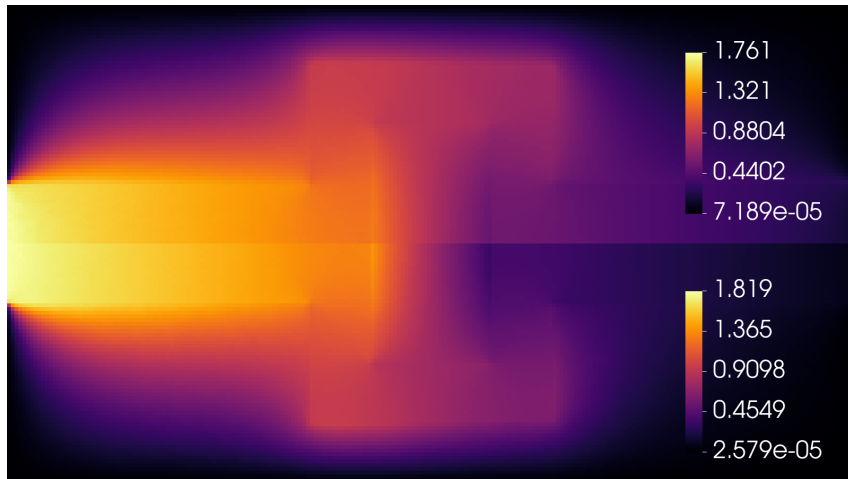
Thus,

$$(M + \Delta t K)\Phi^{n+1} = M\Phi^n + \Delta t f. \quad (19)$$

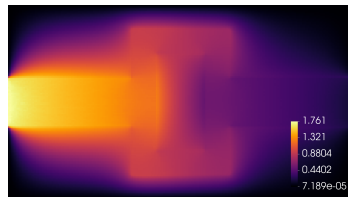
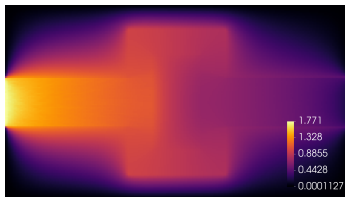
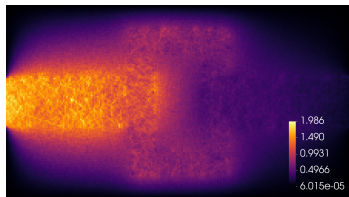
Define parameter α such that,

$$\Delta t = \alpha h^2 \left(\max_{\mathbf{x}} \sigma_t \right). \quad (20)$$

The solution on material interfaces is no longer incorrect.



The generalized variance reduction fixes the noise issue.



The figure of merit is much higher for HSM than for UMC.

	Runtime	Variance	FOM
HSM	0m 22s	$3.66 \cdot 10^{-4}$	124.2
UMC	41m 35s	$5.20 \cdot 10^{-5}$	7.7

Could HSM be competitive with IMD or DDMC?

	HSM	DDMC
Orders of magnitude speedups	✓	✓
No diffusion approximation	✓	×
No phase-space partitioning	✓	×
No issues with unstructured meshes	✓	×
No iteration	×	✓
No linear solver	×	✓
No difference formulation	×	✓
Demonstrated in production calculations	×	✓

Conclusion: the HSM method described in this talk seems useful.

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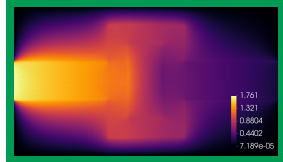
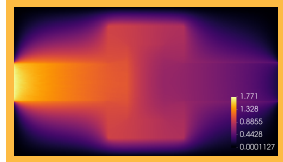
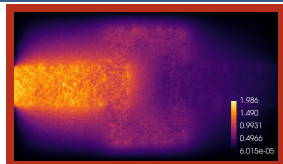
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Conclusion



The future work for HSM is a path to production calculations.

1. time dependence, $\psi(\mathbf{x}, \Omega, t)$
2. frequency dependence, $\psi(\mathbf{x}, \Omega, t, \nu)$
3. 1D Spherical, 2D RZ, and 3D
4. hydrodynamics coupling
5. physical scattering
6. graphics processors
7. spectral line transport
8. high order

Progress Porting LLNL Monte Carlo Transport Codes to the AMD Instinct MI300A APU at 1:25 PM today in Special Session on GPU Computing



llnl.gov/article/52336/llnl-dedicates-el-capitan-ushering-new-era-supercomputing-national-security

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Only 8% of the HSM runtime is in the deterministic component.

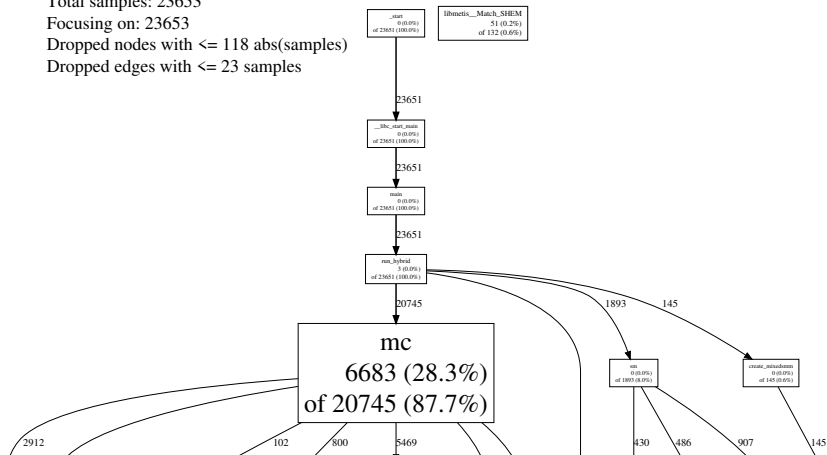
./hyr

Total samples: 23653

Focusing on: 23653

Dropped nodes with ≤ 118 abs(samples)

Dropped edges with ≤ 23 samples



The HSM iteration converges quickly.

```
defaults="
```

```
runvars.convergence_criterion = 0.001;
```

```
runvars.num_bdr_src_particles = 16e6;
```

```
runvars.num_vol_src_particles = 16e6;
```

```
runvars.mesh = [[lcp5.mesh]];
```

```
runvars.output_root_filename = "
```

```
srun -n112 ./hyr lcp.lua -e "$defaults  
[[hsm_lcp_16e6_16e6]]"
```

```
Running lcp5.mesh-16000000-16000000 (mesh-np)
```

```
The total weight is 0.50005
```

```
The total weight is -1.50598
```

```
Cycle 1: Maxres 0.00e+00 0.00e+00 Runtime 8.96e-01 4.78e+00
```

```
The total weight is -1.24976
```

```
Cycle 2: Maxres 4.41e-01 3.04e-01 Runtime 1.08e-01 2.49e+00
```

```
The total weight is -1.15745
```

```
Cycle 3: Maxres 2.13e-01 7.89e-02 Runtime 2.30e-01 8.06e-01
```

```
The total weight is -1.12240
```

```
Cycle 4: Maxres 9.70e-02 2.81e-02 Runtime 9.56e-02 8.10e-01
```

```
The total weight is -1.10688
```

```
Cycle 5: Maxres 3.88e-02 1.14e-02 Runtime 9.18e-02 8.07e-01
```

```
The total weight is -1.09918
```

```
Cycle 6: Maxres 1.71e-02 4.97e-03 Runtime 1.58e-01 8.26e-01
```

```
The total weight is -1.09509
```

```
Cycle 7: Maxres 8.47e-03 2.26e-03 Runtime 1.47e-01 8.03e-01
```

```
The total weight is -1.09284
```

```
Cycle 8: Maxres 4.21e-03 1.06e-03 Runtime 1.53e-01 8.16e-01
```

```
The total weight is -1.09158
```

```
Cycle 9: Maxres 2.11e-03 5.06e-04 Runtime 1.36e-01 8.08e-01
```

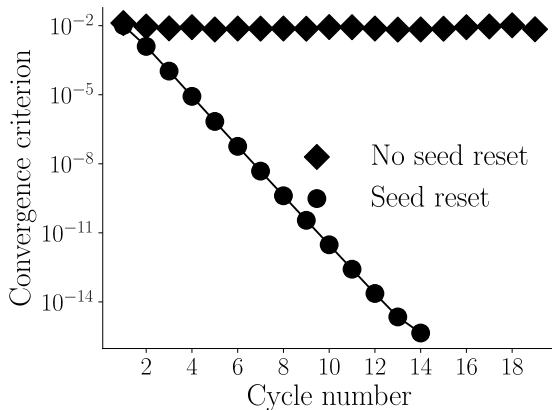
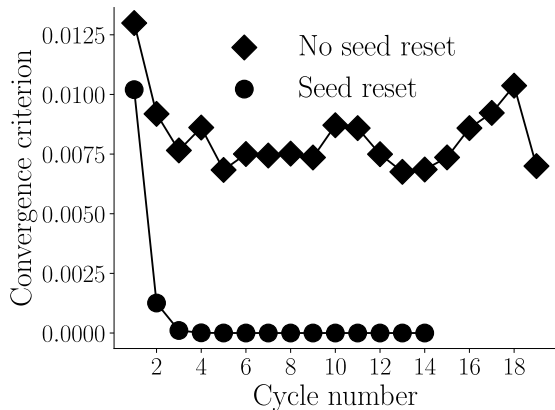
```
The total weight is -1.09085
```

```
Cycle 10: Maxres 1.06e-03 2.48e-04 Runtime 1.39e-01 8.06e-01
```

```
The total weight is -1.09044
```

```
Cycle 11: Maxres 5.31e-04 1.24e-04 Runtime 2.37e-01 8.11e-01
```


Resetting the PRNG seed makes the HSM iteration converge.



We sample volume sources uniformly.

$$\lim_{N \rightarrow \infty} \sum_{i=1}^N w_i = \int_{\mathcal{D}} \int_{\mathbb{S}^2} q \, d\Omega \, d\mathbf{x}, \quad (21a)$$

$$\int_{\mathcal{D}} \int_{\mathbb{S}^2} q \, d\Omega \, d\mathbf{x} \approx \frac{V}{N} \sum_{i=1}^N q(\mathbf{x}_i, \boldsymbol{\Omega}_i), \quad (21b)$$

$$V = \int_{\mathcal{D}} \int_{\mathbb{S}^2} d\Omega \, d\mathbf{x}, \quad \mathbf{x}_i = U(\mathcal{D}), \quad \boldsymbol{\Omega}_i = U(\mathbb{S}^2). \quad (21c)$$

We sample boundary sources uniformly.

$$\lim_{M \rightarrow \infty} \sum_{i=1}^M w_i = \int_{\partial \mathcal{D}} \int_{\boldsymbol{\Omega} \cdot \mathbf{n} < 0} |\boldsymbol{\Omega} \cdot \mathbf{n}| \psi_{\text{inc}} \, d\Omega \, d\mathbf{x}, \quad (22a)$$

$$\int_{\partial \mathcal{D}} \int_{\boldsymbol{\Omega} \cdot \mathbf{n} < 0} |\boldsymbol{\Omega} \cdot \mathbf{n}| \psi_{\text{inc}} \, d\Omega \, d\mathbf{x} \approx \frac{S}{M} \sum_{i=1}^M |\boldsymbol{\Omega}_i \cdot \mathbf{n}| \psi_{\text{inc}}(\mathbf{x}_i, \boldsymbol{\Omega}_i), \quad (22b)$$

$$S = \int_{\partial \mathcal{D}} \int_{\boldsymbol{\Omega} \cdot \mathbf{n} < 0} d\Omega \, d\mathbf{x}, \quad \mathbf{x}_i = U(\partial \mathcal{D}), \quad \boldsymbol{\Omega}_i = U(\mathbb{S}_h). \quad (22c)$$

The iteration converges when the relative difference of successive iterates falls below a user-provided threshold, η .

$$\max_j \left(\frac{|\hat{\phi}_j^{(i-1)} - \hat{\phi}_j^{(i)}|}{\hat{\phi}_j^{(i-1)}} \right) < \eta, \quad j = 1, \dots, |\mathcal{T}|, \quad (23)$$

The error in the two scaling studies is computed differently

$$\text{Element scaling study error} = \frac{||\hat{\phi} - \bar{\phi}||_2}{|\mathcal{T}|}, \quad (24a)$$

$$\text{Particle scaling study error} = ||\hat{\phi} - \bar{\phi}||_2. \quad (24b)$$

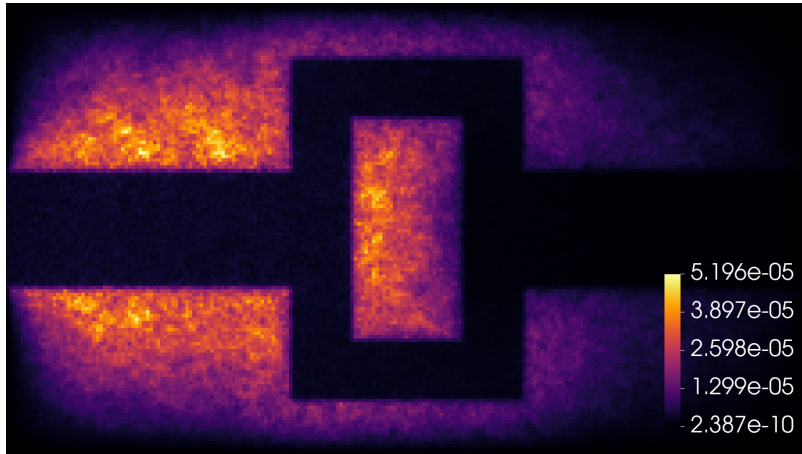
Our code is only 5500 lines, not including dependencies.

Lines of code	Filename
20	wkt_conduction.h
22	wkt_mmsvars.h
26	system.c
37	eval.c
39	wkt_sources.h
57	wkt_runvars.h
481	write.c
746	hyr.c
1621	psmm.cpp
2463	mc.c
5512	total

Version	Library
1.14.3	hdf5
2.33.0	hypre
1.0.0	irep
5.4.6	lua
5.1.0	metis
4.6	mfem
4.40.3	parmetis
5.3.0	superlu



The UMC variance is highest in the optically-thick material.



The HSM variance is highest at the inflow boundary.

